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Requirement Analysis and Specification Document

for

Best Bike Paths

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1 Introduction

1.1 Purpose

Bicycling has become an increasingly popular form of urban transportation and recreation, but cyclists frequently have difficulty finding routes that are safe and comfortable. Available bike path data is fragmented and inconsistent: some paths are well-kept, others are full of potholes or otherwise uncomfortably sloppy to ride with a road bike, or there's traffic constantly interfering. There is no common repository to store and maintain this information centralized, organized, and structured.

The Best Bike Paths (BBP) system fills this gap by offering a centralized data-driven platform for cyclists to document trips and path quality, as well as contribute personal observations to a collective pool available to the public. The BBP includes two modes for gathering data: the manual input mode in which users enter crucial information about their paths travelled and the automatic acquisition capability where GPS (Global Positioning System) and other sensor signals are stored during a ride to recognize suboptimal surfaces. It also includes external weather feeds for better trip reporting and realistic analysis.

By aggregating reports from multiple users and resolving inconsistencies using data freshness and confirmation counts, BBP maintains an up-to-date, reliable inventory of cycling paths to help users plan routes that are both efficient and safer.

1.1.1 Goal

- G1** Allow registered users to record their bike trips and review them later with statistics (distance, duration, average speed, etc.) and related weather information.
- G2** Allow registered users to insert and maintain publishable information about bike paths and obstacles, through manual specification of the travelled paths and reported issues.
- G3** Support automatic detection of possible issues on bike paths during users' trips, and allow users to confirm or correct such detections before they become publishable information.
- G4** Allow any user to find and view suitable bike paths between an origin and a destination, ranked by a path score that considers factors such as length and reported quality (and possibly weather conditions along the paths).
- G5** Merge information collected from multiple users about the same path, resolving conflicts by privileging fresher and more confirmed data so that the resulting path information is reliable.
- G6** Maintain a consistent and up-to-date repository of bike paths and related information that can be used for future analysis and extensions of the service.

1.2 Scope

1.2.1 World Phenomena

The Best Bike Paths (BBP) project takes place in a world in which people use bicycles to move through an urban environment composed of roads and bike paths. In this environment, several relevant phenomena occur independently of the presence of BBP:

WP1 Bikers travelling along physical bike paths and roads

Bikers choose where and when to ride, move along the existing road and bike path network, and complete trips between different locations.

WP2 Evolution of path conditions

The physical and traffic conditions of bike paths change over time because of high traffic, obstacles, maintenance, and other variables.

WP3 Variability of weather conditions

Weather conditions vary by location and time, affecting the comfort, safety, and attractiveness of riding along specific paths.

WP4 Distribution of knowledge about path conditions

Information about the actual conditions and obstacles on bike paths is initially distributed among individual bikers, who may discover issues while riding and may informally share this knowledge (e.g., by word of mouth or private messages) without any centralized support.

WP5 Use of smartphones by bikers

Many bikers carry smartphones capable of determining their position and sensing aspects of their motion (e.g., through GPS and inertial sensors) and can connect to the internet either during or after their trips.

These phenomena characterize the problem world in which BBP is introduced. BBP cannot directly change the physical properties of roads, the weather, or the decisions of bikers about how to ride, but may observe and exploit some of these phenomena through its interaction with users and devices.

1.2.2 Shared Phenomena

BBP is introduced into this world as a software system that interacts with bikers and with external information providers. This interaction gives rise to phenomena that are shared between the world and the system. In particular:

SP1 User authentication and identification (World controlled)

A person provides credentials or profile data to BBP in order to create an account or log in, and BBP recognises that person as a registered user.

SP2 Start and end of trip recording

A registered user initiates and stops the recording of a bike trip through the BBP interface on their device, and BBP records that a trip has started or ended.

SP3 Provision of positional and motion-related data

During a recorded trip (or shortly afterwards), the user's device provides BBP with data related to the position and movement of the bicycle. BBP receives this data and may store or process it.

SP4 Manual contribution and maintenance of path information

A registered user selects path segments they have travelled and provides, updates, or removes publishable information about the status of those segments and about any obstacles encountered. BBP receives these inputs and updates the corresponding stored path information.

SP5 Confirmation or correction of automatically inferred issues

BBP presents a user with candidate issues inferred from previously collected data (e.g., possible obstacles or irregularities on path segments). The user confirms, rejects, or corrects these candidate issues, and BBP updates the corresponding publishable information accordingly.

SP6 Request for a path between two locations

A user (registered or unregistered) specifies an origin and a destination to BBP. BBP interprets this request as a path query.

SP7 Presentation of suggested paths and their properties

In response to a path request, BBP presents one or more bike paths on a map, together with information about their properties (such as length and reported quality) and an indication of their relative ranking. The user views and interprets this information when choosing how to ride.

SP8 Presentation of trip history and statistics

A registered user with a list of their recorded trips and associated statistics can view these records to review past activity.

SP9 Use of external services for map, traffic, and weather information

BBP issues requests to external map/geocoding, traffic, and weather information services and receives responses that provide data about maps, routes, traffic, and weather conditions, which it then uses when enriching trip records or presenting paths to users.

1.2.3 Scope-Purpose Traceability Matrix

Table 2: Scope-purpose traceability matrix for BBP goals.

Goal	Short description	Related world phenomena	Related shared phenomena
G1	Trip recording + stats + weather for registered users	WP1 -Bikers travelling along paths and roads WP3 -Variability of weather conditions WP5 -Use of smartphones by bikers	SP1 -User authentication and identification SP2 -Start and end of trip recording SP3 -Provision of positional and motion-related data SP8 -Presentation of trip history and statistics SP9 -Use of external services for weather information
G2	Manual contribution and maintenance of publishable path/obstacle info	WP1 -Bikers travelling along paths and roads WP2 -Evolution of path conditions WP4 -Distribution of knowledge about path conditions WP5 -Use of smartphones by bikers	SP1 -User authentication and identification SP4 -Manual contribution and maintenance of path information

Goal	Short description	Related world phenomena	Related shared phenomena
G3	Automatic detection of issues + user confirmation/correction	WP1 -Bikers travelling along paths and roads WP2 -Evolution of path conditions WP5 -Use of smartphones by bikers	SP2 -Start and end of trip recording SP3 -Provision of positional and motion-related data SP5 -Confirmation or correction of automatically inferred issues SP9 -Use of external services (if used for context in detection)
G4	Path search between origin and destination + ranked suggestions for any user	WP1 -Bikers travelling along paths and roads WP2 -Evolution of path conditions WP3 -Variability of weather conditions WP4 -Distribution of knowledge about path conditions	SP6 -Request for a path between two locations SP7 -Presentation of suggested paths and their properties SP9 -Use of external map/geocoding and weather services
G5	Merging info from multiple users; privileging fresher/confirmed data	WP2 -Evolution of path conditions WP4 -Distribution of knowledge about path conditions	SP4 -Manual contribution and maintenance of path information SP5 -Confirmation or correction of automatically inferred issues
G6	Maintain a consistent, up-to-date repository of paths for future use	WP2 -Evolution of path conditions WP3 -Variability of weather conditions WP4 -Distribution of knowledge about path conditions	SP4 -Manual contribution and maintenance of path information SP5 -Confirmation/correction of inferred issues SP7 -Presentation of path information to users SP8 -Presentation of trip history and statistics SP9 -Use of external map and weather services

1.3 Definitions, Acronyms, Abbreviations

1.3.1 Definitions

- **Bike Path:** A designated route or lane intended for bicycle travel, which may vary in quality, safety, and accessibility.
- **Trip:** A recorded journey by a user along one or more bike paths, including associated statistics and sensor data.
- **Obstacle:** Any physical or environmental condition (e.g., pothole, debris, construction) that may hinder or endanger cyclists on a bike path.
- **Manual Input Mode:** The feature allowing users to enter information about their trips and encountered obstacles directly into the system.
- **Automatic Acquisition:** The process by which the system collects data from user devices (e.g., GPS, sensors) during a trip to infer path quality and detect issues.

- **Path Score:** A computed value representing the suitability of a bike path, based on factors such as length, reported quality, and weather conditions.
- **Confirmation Count:** The number of users who have confirmed a reported issue or path condition, used to assess data reliability.

1.3.2 Acronyms

- **BBP:** Best Bike Paths
- **GPS:** Global Positioning System
- **API:** Application Programming Interface

1.3.3 Abbreviations

- **G:** Goal
- **WP:** World Phenomena
- **SP:** Shared Phenomena

1.4 Revision History

- Version 1.0: 23/12/2025

1.5 Reference Documents

The assignment for this document and all the information included refer to the following documentation:

- The specification for the 2025/26 Requirement Engineering and Design Project for the Software Engineering II course.
- The slides and materials on the webeep page of the Software Engineering II course.

1.6 Document Structure

This document is organized as follows:

1. **Introduction:** Provides an overview of the project, its goals, and its scope.
2. **Overall Description:** Details the product, including real-world scenarios and a technical analysis of the system.
3. **Specific Requirements:** Lists all requirements necessary to achieve the project goals and their relationship with domain assumptions.
4. **Formal Analysis Using Alloy:** Presents a formal description of selected real-world phenomena using Alloy to highlight inconsistencies between assumptions and modeled phenomena.

5. **Effort Spent:** Summarizes the contributions and time spent by each group member in preparing this document.
6. **References:** Lists documents and tools referenced in the preparation of this document.

2 Overall Description

2.1 Product perspective

2.1.1 Scenarios

This section presents a set of informal scenarios describing how different users can employ the Best Bike Paths (BBP) system under typical circumstances. Each scenario illustrates the actions performed by actors and the corresponding reactions of the system. The scenarios are story-based and independent of implementation or user interface details. They are derived from the goals, world phenomena, and shared phenomena identified in Section 1 and provide the basis for the definition of use cases in later sections.

Scenario S1 — Recording and reviewing a bike trip

A registered user, Alice, goes for a bicycle ride in her neighborhood. Before starting the activity, she logs into BBP and initiates a new trip. While Alice is riding, BBP retrieves position information from her device and records the start time of the trip. At the end of the ride, Alice stops the recording. Later in the day, Alice accesses BBP again, opens the list of recorded trips, selects the trip she completed, and reviews its statistics together with the contextual weather information recorded during the ride.

Involved goals: G1

Main supported functions: F1, F2

Relevant shared phenomena: SP1, SP2, SP3, SP8, SP9

Scenario S2 — Manual contribution of path information Bob, a registered user, regularly rides on a bike path that he notices has recently deteriorated. After completing his ride, Bob logs into BBP and manually selects the streets he travelled along. For one path segment, he indicates that the route *requires maintenance* due to the presence of several potholes. BBP stores Bob's contribution as publishable path information associated with that segment. A few weeks later, after maintenance work has been completed, Bob updates his previous contribution by changing the status of the same path segment to *good condition*.

Involved goals: G2, G6

Main supported functions: F1, F3

Relevant shared phenomena: SP1, SP4

Scenario S3 — Automatic detection and validation of issues Carol is a registered user who enables automatic trip recording before starting her bike ride. During the ride, BBP collects GPS and motion-related data from her device. After the trip ends, BBP analyzes the collected data and detects a possible irregularity on one of the travelled path segments. The system presents Carol with a candidate issue indicating the potential presence of an obstacle. Carol reviews the reported issue and confirms that it corresponds to a real pothole. BBP stores the validated information and makes it available to other users as publishable path information.

Involved goals: G2, G3, G6

Main supported functions: F2, F4

Relevant shared phenomena: SP2, SP3, SP5

Scenario S4 — Searching for a bike path between two locations

David, who may or may not be a registered user, wants to plan a bike ride from his home to a nearby park. He provides BBP with an origin and a destination. The system processes the request, searches for available bike paths between the two locations, and ranks them based on criteria such as route length and

reported conditions. BBP presents several alternative routes together with their scores. David reviews the available options and selects the path he prefers for his journey.

Involved goals: G4

Main supported functions: F5

Relevant shared phenomena: SP6, SP7, SP9

Scenario S5 — Aggregation of information from multiple users Over time, multiple registered users submit information regarding the same bike path segment. Two users report the segment as being in *optimal* condition, while another user confirms an automatically detected issue indicating that maintenance may be required. BBP aggregates the received contributions and resolves the inconsistency by considering the freshness of the data and the number of confirmations associated with each report. As a result, BBP maintains a consistent and up-to-date representation of the path segment, which is used when answering path queries from other users.

Involved goals: G5, G6

Main supported functions: F6

Relevant shared phenomena: SP4, SP5, SP7

2.1.2 Domain Class Diagram

This subsection describes the main concepts that BBP maintains over time and their relationships, independently of user interface or implementation details. The corresponding domain model is shown in Figure 1.

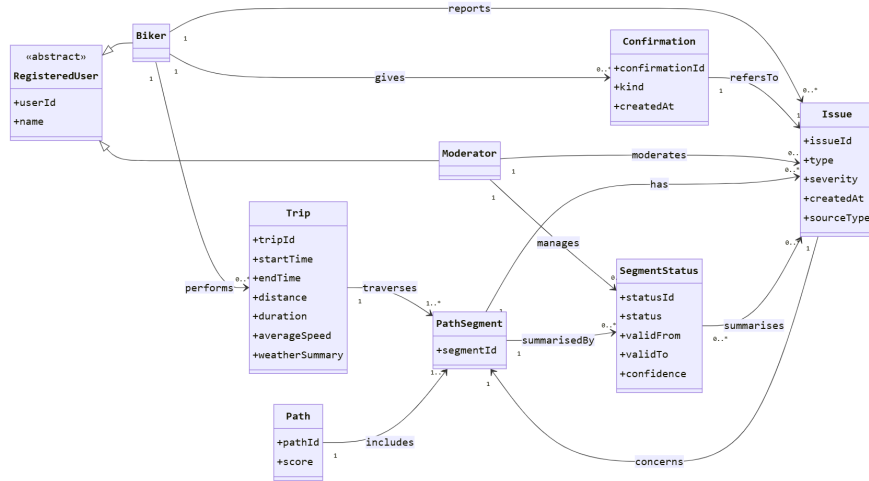


Figure 1: Domain model for Best Bike Paths.

2.1.2.1 Main entities

RegisteredUser, Biker, Moderator. *RegisteredUser* is an abstract concept representing any authenticated user of BBP whose data is stored by the system. It is specialised into:

- **Biker:** a registered user who records trips and contributes path information. In line with the user characteristics, bikers can record trips, manually insert and maintain publishable information about bike paths and obstacles, and confirm or correct automatically inferred issues.
- **Moderator:** a registered user with administrative privileges. Moderators can perform maintenance operations on user-generated content, especially when contributions are clearly abusive or incorrect, and can intervene on the published representation of path information when necessary.

Trip. A *Trip* represents a bike ride actually recorded by a Biker. For each trip, BBP stores identifiers and timestamps and computes basic statistics such as distance, duration, and average speed, together with a summary of the weather experienced during the ride (for example, obtained from the Weather Information Service).

PathSegment and Path. A *PathSegment* is an atomic portion of the bike network (e.g., a street or part of a street between two relevant points). A *Path* is an ordered sequence of *PathSegments* representing a candidate bike route between an origin and a destination. Paths are computed or retrieved by BBP and ranked based on a *score* that takes into account length, reported quality, and possibly weather conditions along the path.

Issue. An *Issue* encapsulates one raw contribution about conditions or obstacles on a single *PathSegment*. In accordance with the functional requirements for manual and automatic contribution of information, Issues may be submitted:

- in **manual mode**, when a Biker selects travelled segments and annotates them with status and obstacles;

- in **automatic mode**, when BBP infers candidate issues from sensor data collected during a trip and then presents them to the user for validation.

Each Issue has a type (e.g., pothole, flood, “good condition”), a severity, a creation time, and a *sourceType* indicating whether it originated manually or automatically.

Confirmation. A *Confirmation* represents the validation action that a Biker performs on an Issue detected by BBP in automatic mode, or on existing information in general. Users can confirm, reject, or refine candidate issues before they become part of the publishable information. Confirmations are stored with a *kind* (e.g., confirm / reject / refine) and a timestamp.

SegmentStatus. *SegmentStatus* models the publishable information about a path segment that BBP maintains and exposes to all users. It captures the current status of a segment (e.g., optimal, medium, sufficient, requires maintenance), its validity interval, and a confidence value reflecting how fresh and well-confirmed the underlying information is. This entity supports the functions related to providing and maintaining publishable information, and to merging information from different users, by providing a consistent, aggregated view of segment conditions used when answering path queries and when presenting detailed path information.

2.1.2.2 Relationships. The most relevant relationships in the domain are the following:

- A **Biker** performs zero or more Trips; each Trip is associated with exactly one Biker.
- Each **Trip** traverses one or more PathSegments, representing the path actually followed during that ride.
- A **Biker** can report zero or more Issues; each Issue concerns exactly one PathSegment.
- A **Biker** can provide zero or more Confirmations, each referring to exactly one Issue.
- A **Moderator** can moderate Issues (for example by flagging or removing clearly incorrect or abusive contributions) and can manage SegmentStatus objects, for instance by overriding or correcting the published representation in exceptional cases.
- A **Path** includes one or more PathSegments and is associated with a path score that is computed using the SegmentStatus of its segments, together with other factors such as length and effectiveness in reaching the destination.
- Each **PathSegment** can have zero or more associated Issues and zero or more SegmentStatus instances over time. At any given moment, at most one SegmentStatus represents the current published status of the segment.
- **SegmentStatus summarises Issues:** for each segment, SegmentStatus objects are conceptually derived by aggregating the Issues reported for that segment and the Confirmations they have received, privileging fresher and more confirmed data. This summarised view is used when ranking paths and when presenting conditions on the map.

Together, these entities and relationships provide a domain-level view that supports all project aspects required for a three-person group: trip recording and review, manual and automatic contribution of publishable path information, confirmation and correction of inferred issues, path search and ranking for any user, and aggregation of information from multiple users to maintain an up-to-date, reliable repository of bike paths.

2.1.3 State Diagrams

Besides the static domain model, some concepts in BBP have a non-trivial lifecycle. They need more emphasis on when they are created and how they evolve over time. State diagrams are used here to describe the possible states of an object and the events that trigger transitions between those states. They help to clarify which operations are allowed in each phase of the lifecycle and how the system guarantees that only consistent, validated information is used when supporting users.

Among the entities defined in the domain model, *Issue* and *SegmentStatus* are the ones whose lifecycle is most critical for the behaviour of BBP. Issues are created in different ways (manual versus automatic), must sometimes be validated by users, and eventually become obsolete or archived. *SegmentStatus* is the publishable summary of all contributions on a segment and is directly used in path ranking; it must react to new information, detect conflicts, and allow moderator intervention. For these reasons, dedicated state diagrams are provided for these two entities (Figures 2 and 3).

2.1.3.1 Issue lifecycle. The state diagram for *Issue* describes how a single Issue (a report about the condition of a segment) evolves from creation to the end of its life. An Issue can be created in two ways: manually by a Biker, or automatically by the system from sensor data collected during a trip. These two modes correspond to two different starting paths in the lifecycle.

Manual path. When a Biker starts filling in a report about a segment, the Issue is conceptually in a “manual draft” state: it exists in the client or in a temporary form, but it is not yet part of the publishable information. When the Biker submits this report, the Issue immediately becomes valid and enters the *Validated* state. From this moment on, it can be used in the aggregation that computes the segment’s published status. If the Biker abandons or cancels the draft before submitting, the Issue is discarded and may go directly to an *Archived* end state without ever becoming visible.

Automatic path. When BBP infers a potential problem from trip data (for example from accelerometer or GPS patterns), it creates an Issue in an *AutoDetected* state. In this state, the Issue represents a candidate problem that has not yet been approved by the user. The Biker is then presented this candidate in the validation interface, moving the Issue into *PendingValidation*. While in *PendingValidation*, the system is waiting for the user’s decision.

From *PendingValidation* there are two main outcomes:

- If the user chooses to confirm the candidate issue (possibly refining its type or severity), the Issue moves to *Validated*. It is now treated like a normal confirmed report and is included in the publishable information.
- If the user rejects the candidate issue, the Issue moves to *Rejected*. Rejected Issues are not used in computing segment status.

Later evolution. A *Validated* Issue may eventually become outdated. For example, the problem might be fixed, or newer reports may indicate that conditions have changed. In this case, the Issue moves from *Validated* to *Obsolete*. Obsolete Issues are information that was once correct but no longer reflects the current state of the segment.

Both *Rejected* and *Obsolete* Issues can then be cleaned up by the system or by a Moderator. This is represented by transitions to an *Archived* state, in which the Issue is kept only for historical or analytical reasons and is no longer taken into account when computing current segment status.

Overall, the Issue state diagram shows that manually submitted reports become *Validated* directly, whereas automatically detected issues must pass through a *PendingValidation* step. It also makes explicit that obsolete or rejected information is eventually archived and does not influence current publishable path information.

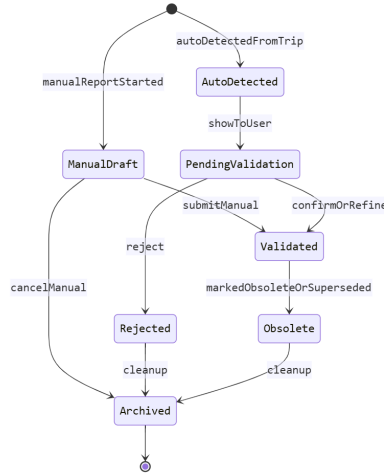


Figure 2: State diagram for the Issue lifecycle.

2.1.3.2 SegmentStatus lifecycle. The state diagram for *SegmentStatus* describes how the published status of a single path segment evolves as BBP receives new information, aggregates reports, and involves Moderators when needed.

A path segment initially has no published status. In this situation, it is considered to be in a *NoStatus* state: BBP has not yet computed any *SegmentStatus* object for that segment. When enough information is available, the system computes a first status for the segment. This may be based on the absence of issues (for example, assigning a default “sufficient” status with low confidence when no problems have been reported) or based on existing Issues and their Confirmations. Once this first status is created, the *SegmentStatus* enters the *Active* state.

The *Active* state represents the current published status of the segment. This is what users see on the map and what the path search function uses to evaluate and rank paths. While in *Active*, the status attributes (such as the qualitative status value and confidence level) may be periodically recomputed as new Issues and Confirmations arrive for that segment. When the new information is broadly consistent with the current status, the aggregation algorithm can update the attributes without changing the state: the *SegmentStatus* remains *Active*, but with refreshed values. This corresponds to automatic merging of contributions that agree with the existing picture.

Sometimes, however, contributions from different bikers might contradict each other in the same time period. For example, some bikers in a given week might report the segment as “good”, but several fresh, high-severity reports indicate that the segment is blocked or dangerous. If the automatic aggregation logic cannot resolve this discrepancy with sufficient confidence, the *SegmentStatus* moves to a *Conflict detected* state. In this state, the system acknowledges that there is a serious disagreement between the existing status and the new evidence.

From the *Conflict detected* state there are two possible directions. If, after receiving more data, the automatic algorithm is able to reconcile the conflict (for instance because one side clearly dominates in terms of recency and confirmations), the *SegmentStatus* can move back to *Active* with updated attributes. This represents conflicts that are resolved automatically by the merging logic. If instead the conflict remains unclear or is explicitly flagged as problematic, the *SegmentStatus* moves to *Under moderation*.

In the *Under moderation* state, a human Moderator reviews the situation. They inspect the underlying Issues and Confirmations, and might have to do site checks, make calls, or check local news to make a decision. If the Moderator decides on a specific status, it is applied and the SegmentStatus moves back to *Active*, now reflecting the status and confidence explicitly set by the Moderator. In more problematic cases—for example, if there is a lot of spam, clearly abusive reports, or the information is too unreliable—the Moderator may decide to temporarily hide the status for the segment. In that case, the SegmentStatus moves from *Under moderation* to *Suspended* (hidden).

When a SegmentStatus is in the *Suspended* state, BBP treats the segment as having no reliable published status. The system might, for example, show a generic “status unavailable” indicator on the map or penalise paths that pass through that segment because its safety is uncertain. At some later time, after further reports or manual cleanup, the Moderator can decide that it is appropriate to show a status again. This is represented by a transition from *Suspended* back to *Active*, possibly with a newly computed or Moderator-assigned status.

Finally, any particular SegmentStatus object can become *Superseded*. When a newer SegmentStatus replaces it as the current Active status for the same segment, or when the Suspended status is permanently retired, it becomes Superseded. In the *Superseded* state, the SegmentStatus is only an archived snapshot and is no longer used when answering path queries. At any given time, each path segment has at most one Active SegmentStatus; other statuses for that segment are either Superseded or in transient states such as *Conflict detected* or *Under moderation*.

Taken together, the SegmentStatus state diagram captures three key aspects of the system: automatic aggregation of many Issues and Confirmations into a single published status, detection and handling of conflicts between new and existing information, and the special role of Moderators in resolving ambiguous situations or temporarily hiding unreliable status information.

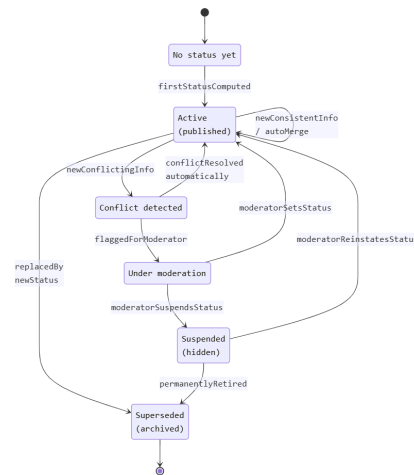


Figure 3: State diagram for the SegmentStatus lifecycle.

2.2 Product Functions

At a high level, the Best Bike Paths (BBP) system provides the following main functions. Each function corresponds to one or more of the goals defined in Section 1.1.

F1 — User Management

BBP manages registered user accounts. It allows people to create an account, authenticate,

and access functionalities that are reserved to registered users (such as trip recording and contribution of publishable path information).

(mainly supports G1, G2, G3)

F2 — Trip Recording and Review

BBP allows registered users to record their bike trips and to review them afterwards. For each recorded trip, the system stores basic information such as the path followed and the time interval, and it computes and presents statistics such as distance, duration, and average speed, together with contextual information such as the weather experienced during the trip.

(mainly supports G1)

F3 — Manual Contribution and Maintenance of Path Information

BBP allows registered users to manually provide and maintain publishable information about bike paths. Users can indicate path segments they have ridden and annotate them with information about status and obstacles. Users can later modify or remove their own contributions when conditions change.

(mainly supports G2, G6)

F4 — Support for Automatic Detection and Validation of Issues

BBP supports the automatic detection of possible issues on bike paths during users' trips and their subsequent validation. Based on sensor data collected while a user is riding, the system can infer candidate issues on path segments and present them to the user, who can confirm, reject, or refine them. Only validated information becomes part of the publishable path information.

(mainly supports G2, G3, G6)

F5 — Path Search and Ranked Presentation

BBP allows any user to request one or more bike paths between a given origin and destination. The system computes or retrieves suitable paths, presents them on a map, and provides a ranking based on a path score that considers factors such as length, reported quality, and weather conditions along the paths.

(mainly supports G4, G6)

F6 — Aggregation and Maintenance of Path Knowledge

BBP aggregates information from multiple users about the same path segments and maintains a consistent, up-to-date view of their status and obstacles. When contributions conflict, the system resolves them by privileging fresher and more validated data. This aggregated view is used when answering path queries and when presenting detailed path information to users.

(mainly supports G5, G6)

2.3 User Characteristics

BBP is intended for the following categories of users and external actors.

2.3.1 End Users

2.3.1.1 Unregistered User

A person who accesses BBP without having created an account, or without logging in. Unregistered

users can:

- Request bike paths between an origin and a destination.
- View suggested paths and their basic properties on a map.

They are assumed to have basic familiarity with mobile or web applications and with map-based interfaces, but they do not need technical expertise.

2.3.1.2 Registered User (Biker)

A user who has created an account and is authenticated in BBP. In addition to the capabilities of unregistered users, registered users can:

- Record their own bike trips and later review trip history and statistics.
- Manually insert, update, and remove publishable information about bike paths and obstacles.
- Confirm, reject, or refine issues that BBP has inferred from data collected during their trips.

Registered users are assumed to be regular or occasional bicycle riders, carrying a smartphone while riding and having at least intermittent internet connectivity. They are not required to have technical knowledge beyond normal use of smartphone applications.

2.3.1.3 System Administrator / Moderator

A user with administrative privileges on BBP. Administrators can:

- Perform maintenance operations on the system.
- Access diagnostic information.
- Moderate user-generated content when needed, for example by handling clearly abusive or obviously incorrect reports.

Administrators are assumed to have technical expertise in system administration and are not representative of typical end users.

2.3.2 External System Actors

These actors provide data or services that BBP relies on while processing user requests.

2.3.2.1 Weather Information Service

An external service that provides weather data for specific locations and times. BBP queries this service in order to:

- Associate weather information with recorded trips.
- Consider expected weather along candidate paths when available.

2.3.2.2 Map / Geocoding / Routing Service

An external service that provides map data, geocoding, and, where available, routing information. BBP uses this service in order to:

- Represent paths and locations on a map.
- Translate user-specified locations (e.g., addresses) into coordinates.
- Compute or retrieve candidate bike paths between an origin and a destination.

2.4 Assumptions, Dependencies and Constraints

2.4.1 Domain Assumptions

This section lists the main domain assumptions and environmental constraints that affect the requirements of BBP but are not under the direct control of the system. They are used together with the requirements to achieve the goals identified in Section 1.1.

DA1 Availability of suitable devices for end users

Registered users access BBP from personal devices (typically smartphones) equipped with:

- a GPS receiver;
- motion-related sensors (e.g., accelerometer, gyroscope);
- display and input mechanisms that support interaction with the BBP user interface.

The device operating system permits the BBP client to read these sensors in line with the platform security and privacy policies.

DA2 Network connectivity

Devices used by registered and unregistered users have at least intermittent connectivity to the BBP back end. Continuous connectivity during an entire trip is not guaranteed, yet data recorded locally is assumed to reach BBP within a reasonable time after the ride.

DA3 Availability and quality of external services (WIS, MGRS)

The Weather Information Service (WIS) and the Map/Geocoding/Routing Service (MGRS) remain available and provide sufficiently accurate, up-to-date data for BBP. Temporary outages or inaccuracies can degrade functionality (e.g., weather enrichment or path computation), but overall availability is assumed.

DA4 Existence and stability of the road and bike path network

BBP operates in areas with a road and bike path network that can be represented by MGRS. Changes to the network (such as new roads or closures) are eventually reflected in the data provided by MGRS, even if the updates are not immediate.

DA5 Reasonable behaviour of users

Registered users are expected to act in good faith when they:

- record trips,
- manually contribute information about paths and obstacles,
- confirm or correct automatically inferred issues.

Occasional mistakes are anticipated, but systematic malicious behaviour (e.g., spamming false reports) is exceptional and only partially mitigated by aggregation and moderation mechanisms.

DA6 Non-critical usage context and user responsibility

BBP supports planning and review of bike trips but is not safety-critical. Users remain responsible for their own safety and compliance with traffic laws and regulations. The application cannot detect real-time path conditions and therefore cannot be held liable for accidents or damages.

DA7 Regulatory and platform constraints on data handling

The collection, storage, and processing of trip data and path information must comply with applicable data-protection and privacy regulations, as well as with the usage policies of the platforms on which BBP runs and of the external services it uses. These requirements may constrain, for example, how long detailed trip data can be retained, how it can be shared, or which consents must be obtained from users.

2.5 Functional Requirements

2.5.1 Use Case Diagram

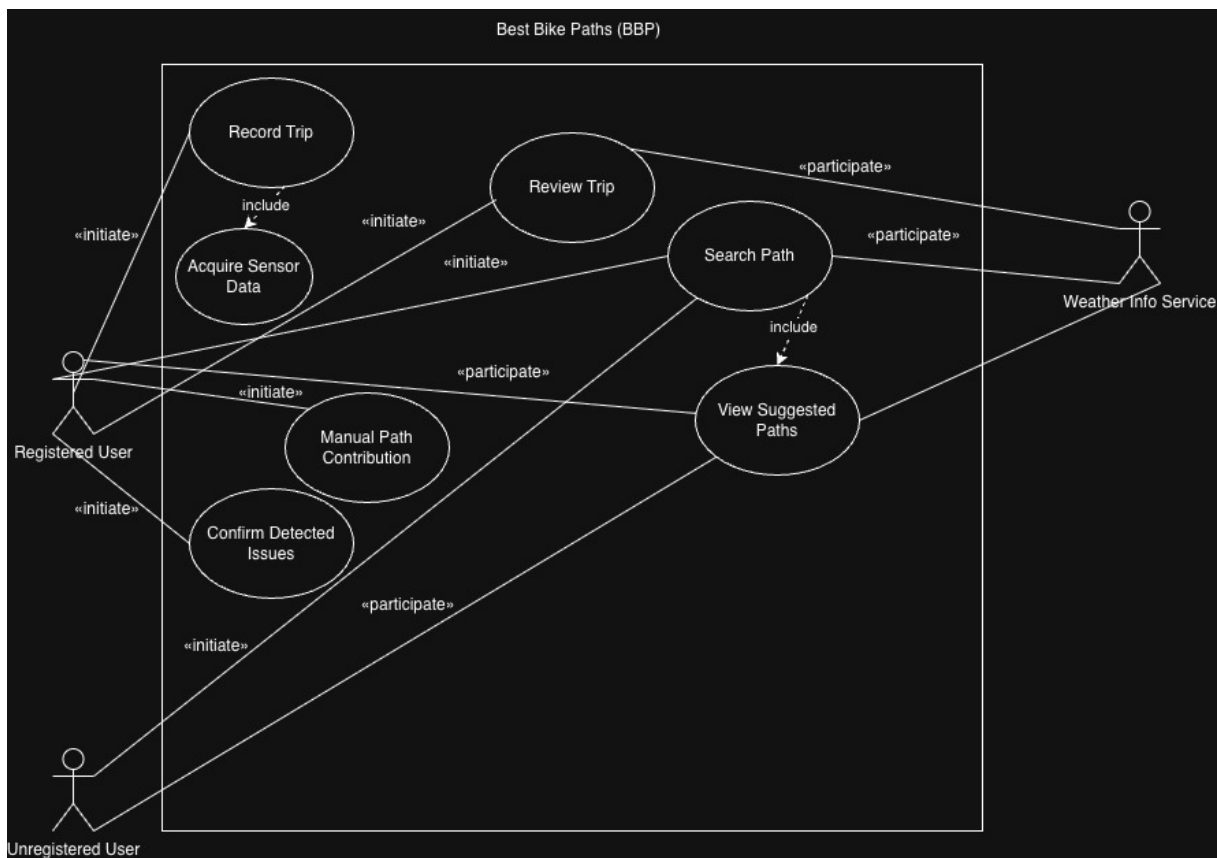


Figure 4: Use Case Diagram for the Best Bike Paths (BBP) system

2.5.2 Use Cases

Every use case is provided with a sequence diagram which specifies the dynamic behavior of actors and system components when they interact with each other. The detailed descriptions of included use cases such as UC1.1 and UC6 have been presented in both written form and in referenced sequence diagrams.

2.5.2.1 [UC1] Record Trip

Name	Record Trip
Actors	Registered User
Entry Condition	The user is logged into BBP on a device capable of recording trips
Event Flow	(a) The user opens the Trips section (b) The user selects the option to start recording a new trip (c) The system records the start time and creates a new trip record in recording state (d) While the trip is ongoing, the system periodically collects positional and motion-related data from the user's device (e) At the end of the ride, the user selects the option to stop recording the trip (f) The system finalizes the trip record, stores the collected data, and marks the trip as completed
Exit Condition	A completed trip is stored in the system with associated raw data and can later be processed to compute statistics and enrich it with contextual information
Exceptions	(d) The device is temporarily unable to provide positional data. The system continues recording but marks affected segments as incomplete (e) The application is closed unexpectedly before the user stops the trip. The system attempts to recover the latest partial recording and may flag the trip as incomplete

2.5.2.2 [UC1.1] Acquire Sensor Data (included in UC1)

Name	Acquire Sensor Data
Actors	Registered User
Entry Condition	UC1 Record Trip has started and the trip is in recording state
Event Flow	(a) At regular intervals, the system requests GPS position from the device (b) At the same time, the system acquires accelerometer and gyroscope samples describing the motion of the device (c) The system associates sensor samples with the current trip and orders them over time (d) The system stores the collected samples for later analysis
Exit Condition	A consistent set of time-stamped sensor readings is stored and associated with the corresponding trip
Exception	(a) Some sensor readings are unavailable or out of range. The system skips those samples and continues acquiring subsequent ones

2.5.2.3 [UC2] Review Trip

Name	Review Trip
Actors	Registered User
Entry Condition	The user is logged into BBP and has at least one recorded trip
Event Flow	(a) The user opens the Trips section (b) The system displays the list of previously recorded trips with basic information (c) The user selects one trip from the list (d) The system displays detailed information about the selected trip, including reconstructed path, statistics such as distance, duration, average speed, and, when available, contextual weather information for the time and area of the trip
Exit Condition	The user successfully views the details and statistics of a selected trip
Exception	(b) No trips are available. The system informs the user and may offer to start a new recording

2.5.2.4 [UC3] Manual Path Contribution

Name	Manual Path Contribution
Actors	Registered User
Entry Condition	The user is logged into BBP and wants to create or update publishable information about bike paths they have travelled
Event Flow	(a) The user opens the section dedicated to manual path contributions (b) The system allows the user to select one or more path segments (c) For each selected segment, the user provides or updates information such as path status and any relevant obstacles (d) The user confirms the submission or modification (e) The system validates the input and updates the stored publishable path information associated with the selected segments
Exit Condition	The system stores new or updated publishable information for the selected path segments, associated with the contributing user
Exceptions	(c) Required information is missing. The system displays an error message and requests the user to complete the missing fields (e) The selected path cannot be resolved. The system warns the user and discards or partially applies the modification

2.5.2.5 [UC4] Confirm Detected Issues

Name	Confirm Detected Issues
Actors	Registered User
Entry Condition	The user is logged into BBP and there exist one or more candidate issues inferred automatically from their previously recorded trips
Event Flow	(a) The user opens the section showing detected but unconfirmed issues (b) The system displays a list of candidate issues, each associated with a path segment and basic contextual information (c) For each candidate issue, the user chooses one of the following: (i) confirm the issue as valid (ii) reject it as a false positive (iii) correct it (d) The system updates the corresponding publishable information for the path segment according to the user's decision
Exit Condition	All processed candidate issues are either confirmed, rejected, or corrected, and the corresponding publishable path information is updated
Exception	(b) No pending issues are available. The system informs the user and returns to the previous view

2.5.2.6 [UC5] Search Path

Name	Search Path
Actors	Registered User, Unregistered User, Weather Information Service
Entry Condition	A user (registered or unregistered) accesses BBP and wishes to find bike paths between two locations
Event Flow	(a) The user opens the Path Search (b) The user specifies an origin and a destination, optionally adding basic preferences (c) The system validates the input and interacts with the external weather or map information service to resolve locations and retrieve contextual data needed for scoring (d) The system identifies candidate bike paths between the origin and the destination, based on stored path information and constraints (e) The system invokes the View Suggested Paths use case (included) to present the resulting alternatives to the user
Exit Condition	A set of candidate bike paths between the specified origin and destination has been computed and is ready to be presented to the user
Exceptions	(c) The destination cannot be resolved. The system notifies the user and asks for correction (d) No feasible path is found. The system informs the user that no suitable bike paths are currently available for the requested locations

2.5.2.7 [UC6] View Suggested Paths

Name	View Suggested Paths
Actors	Registered User, Unregistered User, Weather Information Service
Entry Condition	The system has computed one or more candidate bike paths between an origin and a destination
Event Flow	(a) The system displays the set of candidate paths on a map or in a list, showing for each path: (i) its approximate route (ii) length and other effectiveness metrics (iii) aggregated quality status derived from publishable path information (iv) when available, relevant weather information along the path (b) The system orders the paths according to a path score that combines effectiveness and reported quality (c) The user browses the alternatives and may select one path to inspect in more detail (d) The system shows additional details for the selected path
Exit Condition	The user has been able to inspect and compare the suggested bike paths. The user may select one as the preferred route for their ride
Exception	(a) Only one feasible path is available.

2.5.3 Sequence Diagrams

2.5.3.1 [UC1] Record Trip



Figure 5: Sequence diagram for [UC1] Record Trip

2.5.3.2 [UC1.1] Acquire Sensor Data (included in UC1)

UC1.1 - Acquire Sensor Data

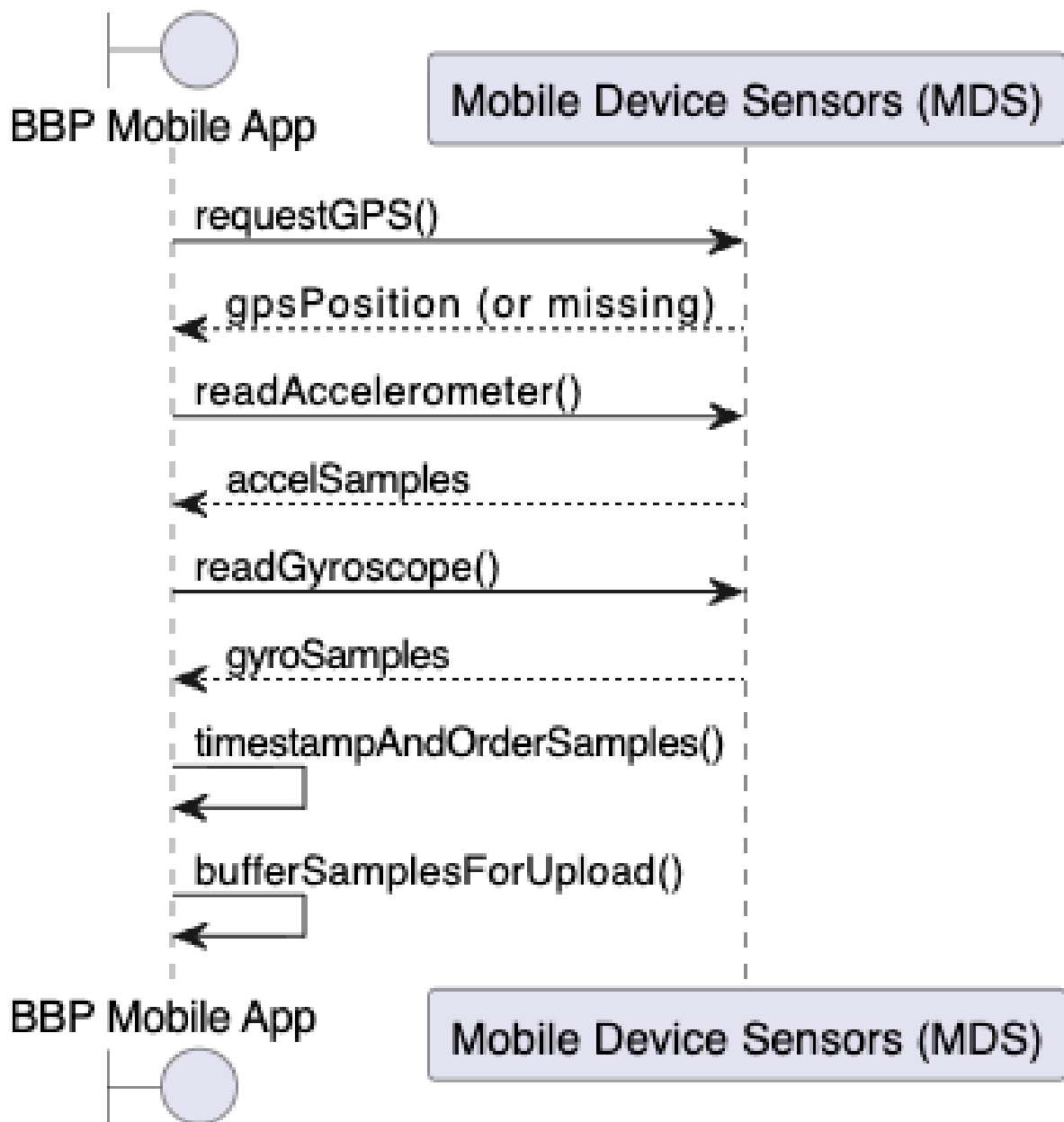


Figure 6: Sequence diagram for [UC1.1] Acquire Sensor Data

2.5.3.3 [UC2] Review Trip

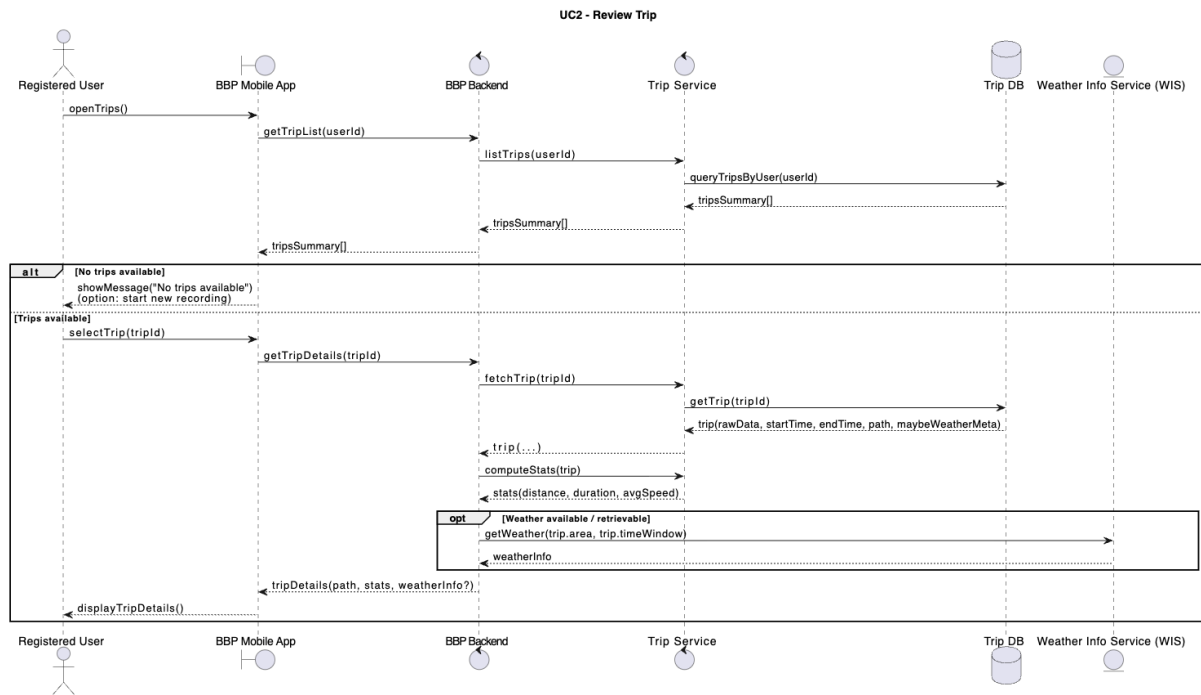


Figure 7: Sequence diagram for [UC2] Review Trip

2.5.3.4 [UC3] Manual Path Contribution

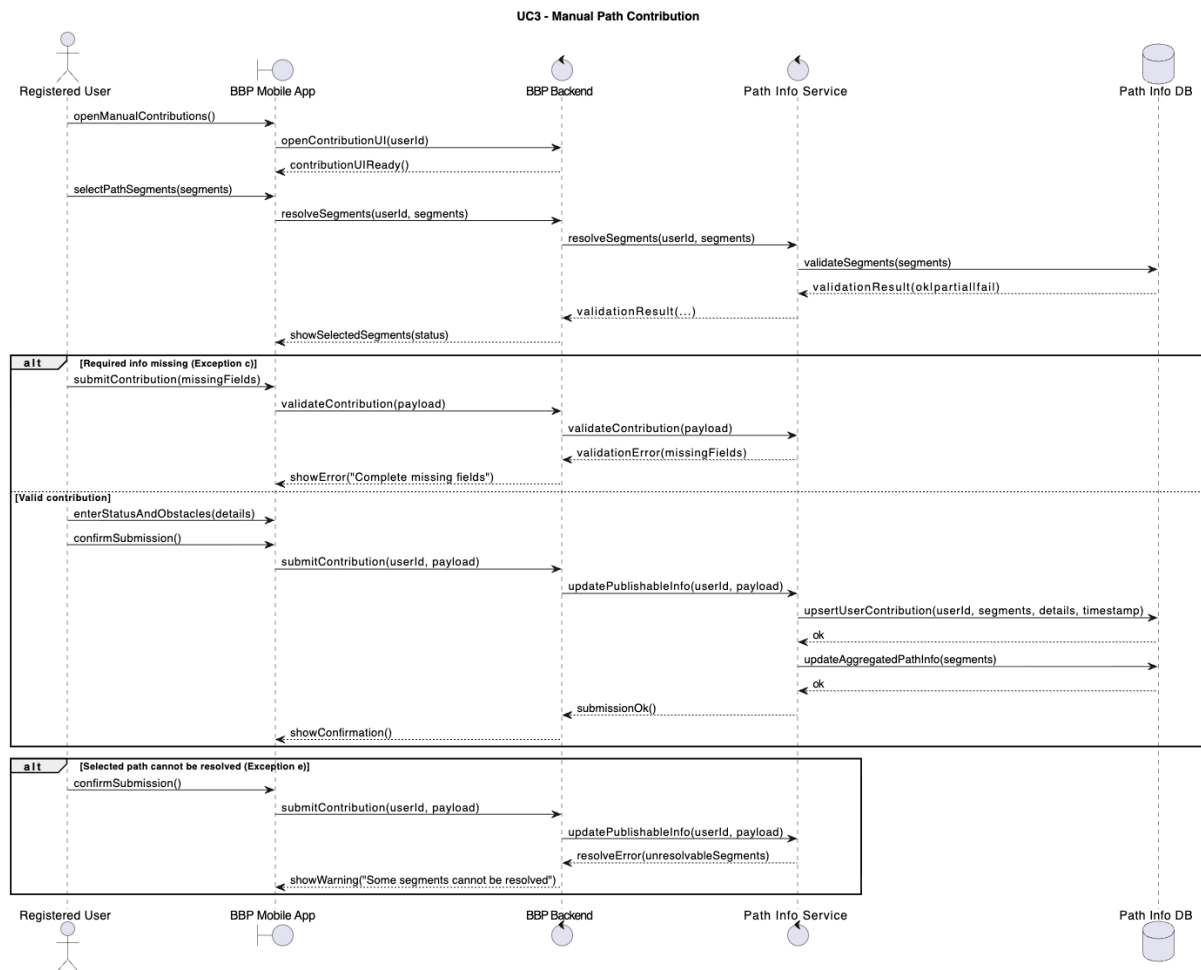


Figure 8: Sequence diagram for [UC3] Manual Path Contribution

2.5.3.5 [UC4] Confirm Detected Issues

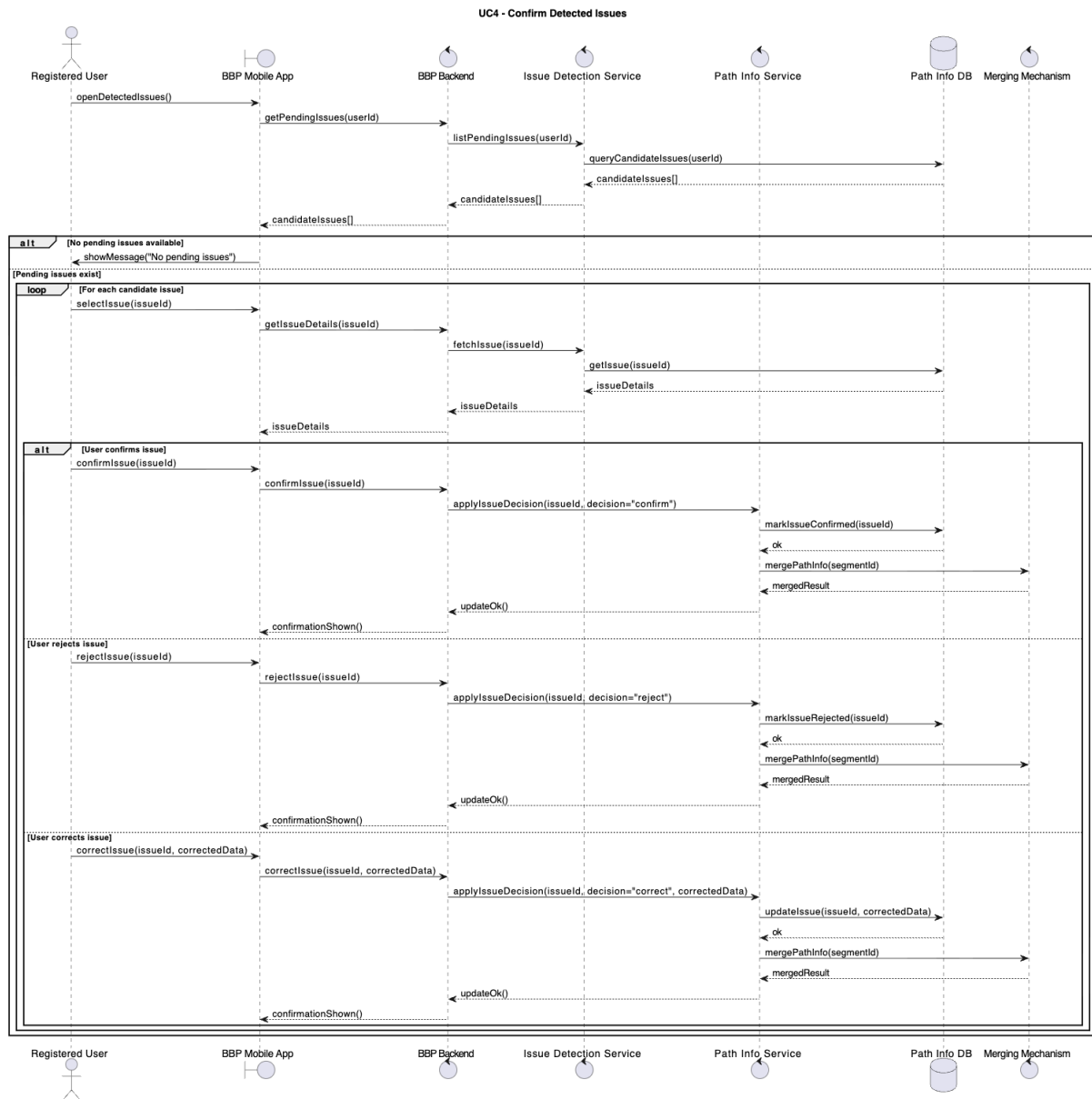


Figure 9: Sequence diagram for [UC4] Confirm Detected Issues

2.5.3.6 [UC5] Search Path

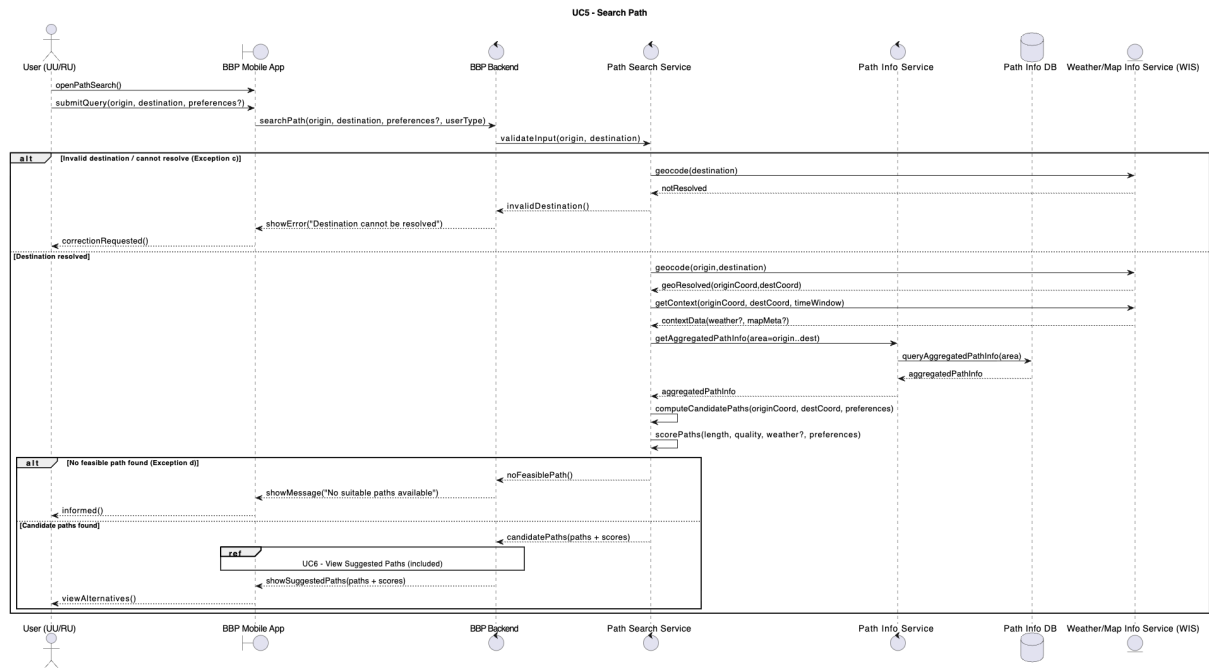


Figure 10: Sequence diagram for [UC5] Search Path

2.5.3.7 [UC6] View Suggested Paths



Figure 11: Sequence diagram for [UC6] View Suggested Paths

2.5.4 Requirement Mapping

G1: Registered users can record bike trips, review trip history, and visualize statistics enriched with contextual information such as weather.
Requirements: • R1: The system allows registered users to authenticate and be identified. • R2: The system allows users to start and stop the recording of a bike trip. • R3: The system acquires positional and motion-related data from mobile devices. • R4: The system stores recorded trips and allows users to review trip history. • R5: The system computes and displays statistics related to recorded trips. • R6: The system integrates external services to retrieve contextual weather information. Domain Assumptions: • D1: Users carry smartphones equipped with GPS and motion sensors. • D2: Users correctly start and stop trip recordings. • D3: External weather services provide reliable data. • D4: Users have a stable internet connection during synchronization phases.

Table 3: Requirement mapping for Goal G1

G2: Registered users can manually contribute and maintain publishable information about bike paths and obstacles.
Requirements: • R1: The system allows registered users to authenticate and be identified. • R7: The system allows users to manually select path segments. • R8: The system allows users to add or update information about path conditions and obstacles. • R9: The system validates and stores publishable path information. Domain Assumptions: • D5: Users provide accurate and truthful information about path conditions. • D6: Users are familiar with the paths they contribute information about.

Table 4: Requirement mapping for Goal G2

G3: The system automatically detects potential issues on paths and allows users to confirm, reject, or correct them.
Requirements: • R2: The system records trips and collects sensor data. • R3: The system analyzes recorded data to infer potential issues. • R10: The system presents detected issues to users for validation. • R11: The system allows users to confirm, reject, or correct detected issues. • R12: The system updates publishable path information based on user decisions. Domain Assumptions: • D7: Automatically detected issues may include false positives. • D8: Users are willing and able to validate detected issues.

Table 5: Requirement mapping for Goal G3

G4: Users can search for bike paths between two locations and receive ranked suggestions based on effectiveness and quality.
Requirements: • R13: The system allows users to specify origin and destination locations. • R14: The system computes candidate bike paths between locations. • R15: The system ranks paths based on length, quality, and contextual data. • R16: The system displays suggested paths and their properties. • R6: The system integrates external map and weather services. Domain Assumptions: • D9: External map and geocoding services return correct results. • D3: Weather services provide reliable contextual information.

Table 6: Requirement mapping for Goal G4

G5: The system merges information from multiple users, privileging fresher and confirmed data. Requirements: • R9: The system maintains publishable path information. • R11: The system updates path data based on user confirmations or corrections. • R17: The system merges contributions and resolves conflicts. Domain Assumptions: • D10: More recent and confirmed data better reflects current path conditions.

Table 7: Requirement mapping for Goal G5

G6: The system maintains a consistent and up-to-date repository of bike paths for future use. Requirements: • R9: The system stores publishable path information. • R12: The system applies updates derived from user contributions and confirmations. • R16: The system presents up-to-date path information to users. Domain Assumptions: • D11: Path conditions evolve over time and require continuous updates.

Table 8: Requirement mapping for Goal G6

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

This subsection presents a high-level illustration of the core user interfaces of Best Bike Paths (BBP). The goal is to give a specific visual design and clarify the main interaction flows. The subsection explores the information that must be presented, and how these interfaces support the goals and product functions defined in Sections 1.1 and 2.2.

Unless otherwise stated, we assume a smartphone client application used in portrait orientation. Both unregistered users and registered users (Bikers and Moderators) interact with BBP through map-based screens with a consistent layout: a top bar with navigation and context, a central map or content area, and bottom panels for actions and lists. This is consistent with the domain assumptions on user devices (DA1) and connectivity (DA2).

3.1.1.1 Home / Map and Path Search

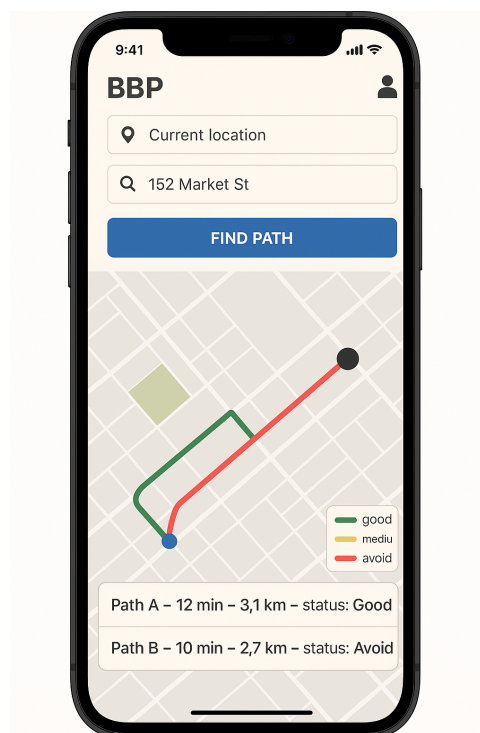


Figure 12: Conceptual home / map screen for path search.

The home screen (Figure 12) is the main entry point for both unregistered users and Bikers when searching for paths (F5). It corresponds to Scenario S4 and to shared phenomena SP6–SP7.

At the top, a simple form allows the user to specify an *origin* and a *destination*. By default, the origin field is pre-filled with the current location, while the destination field accepts a free-text address or point of interest. A prominent “Find path” button triggers the path query.

The central area shows a map with the user position and the candidate paths between the chosen origin and destination. At least two alternative paths are shown, using different colours to represent their overall status (e.g., green for good, yellow for medium, red for avoid). A small legend clarifies the meaning of these colours. A bottom sheet lists the candidate routes with their main properties (estimated time, distance, summary status). Selecting a path in the list highlights it on the map. This directly supports requirements RE-F5–1 to RE-F5–5.

3.1.1.2 Trip Recording screen

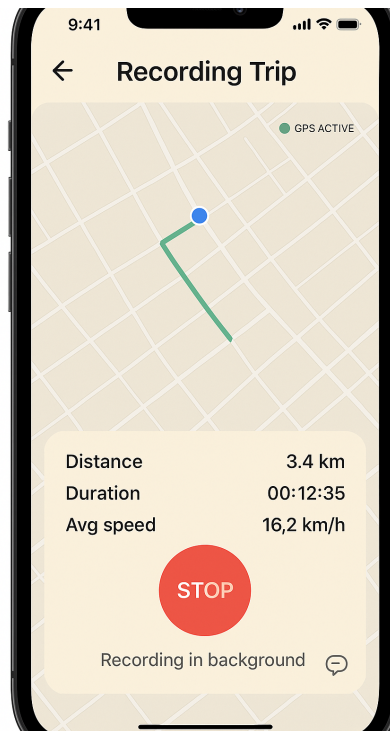


Figure 13: Conceptual trip-recording screen for registered users.

The trip recording screen (Figure 13) is available only to registered users (Bikers). It supports Scenario S1 and function F2.

The top bar indicates the current mode (e.g., “Recording Trip”). The central map shows the current GPS position and the portion of the path already travelled during the active trip. A small indicator (e.g., “GPS active”) provides feedback on sensor availability (SP2–SP3).

A stats panel above the bottom edge displays key trip statistics updated in real time: distance, elapsed time, and average speed. A large, high-contrast button in the panel controls recording: it appears as “Start” when no recording is active, and as “Stop” (or “Pause”) while recording is ongoing. A short status message (such as “Recording in background”) confirms the current state. An optional “Add note / mark issue” control lets the user drop a marker on the current segment to be refined later in the contribution screen. Together, these elements support requirements RE-F2–1 to RE-F2–6.

3.1.1.3 Manual Contribution / Path Information

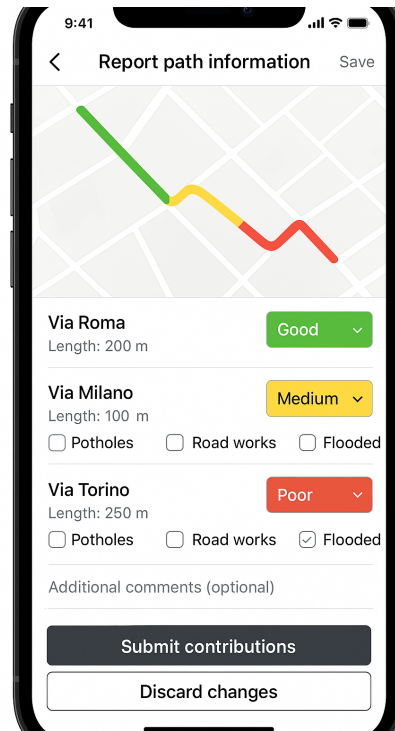


Figure 14: Conceptual manual contribution screen.

The manual contribution screen (Figure 14) is used by Bikers to insert and update publishable information about the path segments they have travelled (Scenario S2, function F3, shared phenomenon SP4).

At the top, the map displays the most recently travelled path. The path is broken into separate segments; each selected segment is shown in a colour that reflects its chosen status (e.g., green for good, yellow for medium, red for poor/avoid). Below the map, a list contains one entry per selected segment. Each entry shows:

- the street name and approximate length;
- a drop-down menu to choose the segment status (Good / Medium / Poor / Avoid);
- checkboxes to indicate specific obstacles (e.g., potholes, road works, flooded);
- an optional text field for comments at the bottom of the list.

A primary “Submit contributions” button confirms the edits and stores them as publishable information (RE-F3–2, RE-F3–4), while “Discard changes” allows the user to cancel the operation without updating the stored data (RE-F3–3).

3.1.1.4 Validation of Automatically Detected Issues

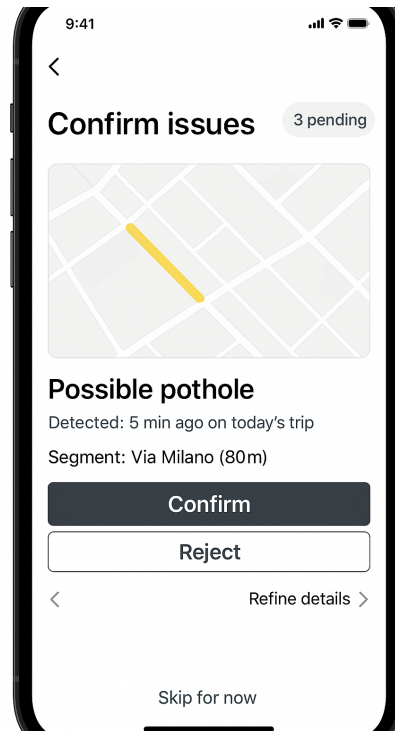


Figure 15: Conceptual screen for validating automatically detected issues.

The issue confirmation screen (Figure 15) is presented to Bikers when BBP has inferred candidate Issues from sensor data collected during a recorded trip (Scenario S3, function F4, shared phenomenon SP5).

At the top, the screen shows the title (e.g., “Confirm issues”) and the number of pending candidate issues. The central panel displays one Issue at a time as a card containing:

- a mini-map with the affected segment highlighted;
- a short textual description (issue type, detection time on the current trip, segment name and length).

Below the card, two large buttons allow the user to *Confirm* or *Reject* the candidate Issue. A secondary text “Refine details” opens a compact form in case the biker wants to edit before confirming. A “Skip for now” action leaves the Issue in the *PendingValidation* state. Confirmed or refined Issues are stored as validated information, while rejected Issues are discarded, in line with requirements RE-F4–2 to RE-F4–5.

3.1.1.5 Moderator Review screen

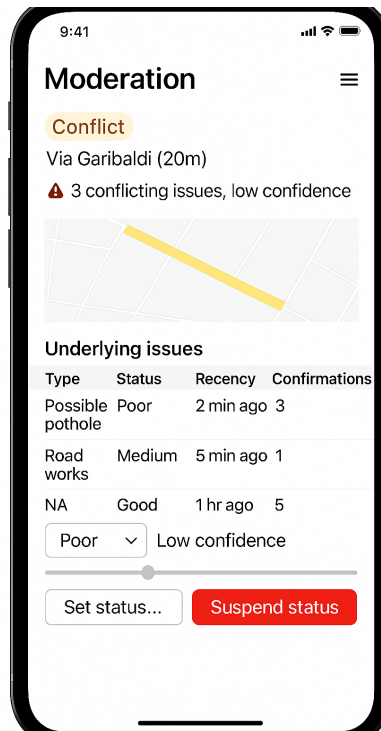


Figure 16: Conceptual moderation screen for resolving conflicts.

The moderation screen (Figure 16) is intended for Moderators and supports the conflict-resolution and aggregation behaviour described in Scenario S5, function F6, and in the SegmentStatus state diagram.

At the top, a summary shows the current problem segment (e.g., “Via Garibaldi (20m)”) together with a chip indicating whether its SegmentStatus is in *Conflict* or *Suspended*. A short line summarises the situation (e.g., “3 conflicting issues, low confidence”). Below, a map highlights the affected segment.

The “Underlying issues” table lists the Issues that contributed to the conflict, with columns for type, status (e.g., Good/Bad), recency (time since the report), and number of confirmations. This directly addresses the aggregation and conflict-detection logic (F6).

At the bottom of the screen, some inputs allows the Moderator to:

- choose a new published status for the segment from a drop-down menu (Good / Medium / Poor / Avoid);
- adjust a confidence value via a slider;
- apply the chosen status (“Set status” button);
- temporarily hide the status by pressing a prominent “Suspend status” button, moving the SegmentStatus into the Suspended state.

These actions correspond to the transitions between *Active*, *Conflict detected*, *Under moderation* and *Suspended* in the SegmentStatus lifecycle.

3.1.1.6 Consistency with requirements

The five screens described above collectively cover all the main interactions defined in the RASD: trip

recording and review (G1, F2), manual contribution and maintenance of path information (G2, F3), validation of automatically detected issues (G3, F4), path search and ranked presentation for any user (G4, F5), and aggregation and moderation of contributions (G5–G6, F6). They are intended as illustrative examples; the final implementation may vary in appearance as long as these interaction patterns and information needs are preserved.

3.1.2 Hardware interfaces

BBP is designed primarily for smartphones carried by bikers and for server-side hardware operated by the service provider.

3.1.2.1 Client devices.

- **Smartphone:** modern Android or iOS device.
- **Location and motion sensors:** integrated GPS receiver and inertial sensors inside the mobile, usually an accelerometer and a gyroscope. They help to measure position and motion data during trips.
- **Network connectivity:** Wi-Fi or mobile data (3G/4G/5G) to communicate with the BBP backend and external services.

The smartphone hardware is controlled by the user and its operating system. BBP can only access sensors and networks that have been allowed by the users and their devices' OSes.

3.1.2.2 Server-side hardware. BBP backend services utilize general-purpose server hardware. No assumptions are made on the specific CPU, memory, or storage configuration other than being sufficient to:

- handle concurrent requests from client devices;
- store and retrieve trip records, path information, and user accounts;
- communicate with external map, traffic, and weather services.

Details of server provisioning and scaling are considered design and deployment concerns and are outside the scope of this RASD.

3.1.3 Software interfaces

BBP interacts with many external software systems, platforms and services.

3.1.3.1 Map / geocoding / routing service. An external map service provides:

- digital map tiles for visualization in the client UI;
- geocoding and reverse-geocoding of addresses and points of interest;
- routing between origin and destination, including alternative paths.

The interface is a web API invoked by the BBP backend; requests contain origin/destination coordinates or textual addresses, and responses contain route geometries, distances, and metadata.

3.1.3.2 Weather information service. A weather service provides current and recent weather conditions for given locations and times. BBP uses this interface to:

- enrich trip records with observed weather conditions;
- optionally influence path ranking based on adverse weather.

The interface is a web API that accepts geographic coordinates and timestamps and returns structured weather data (e.g., temperature, precipitation, wind).

3.1.3.3 Mobile platform services. On client devices, BBP relies on operating system services exposed through standard SDKs:

- location services (to access GPS position with user consent);
- sensor APIs (to read accelerometer and related motion data);
- network APIs (to perform secure HTTP requests to the BBP backend).

These interfaces are platform-dependent but are logically equivalent across supported mobile operating systems.

3.1.4 Communication interfaces

BBP uses standard internet protocols to exchange data between client applications, backend services, and external providers.

3.1.4.1 Client-server communication.

- **Protocol:** HTTPS over TCP/IP.
- **Interaction style:** REST-like web API exposed by the BBP backend.
- **Data format:** produce structured JSON payloads for requests and responses (e.g., trip uploads, path queries, user authentication, management of path information and issues).

All communication between client devices and the BBP backend must be encrypted using TLS. Authentication tokens (for registered users) are transported in HTTP headers or equivalent secure mechanisms.

3.1.4.2 Server-external service communication. The BBP backend communicates with external map, traffic, and weather services through:

- HTTPS-based web APIs provided by those services;
- JSON or comparable structured formats, as defined by the respective providers;
- API keys or similar credentials managed on the server side and never exposed to client devices.

3.1.4.3 Offline and intermittent connectivity. If the client device is temporarily offline, trip data and issue reports may be cached locally on the device and sent to the backend when connectivity is restored. The precise buffering strategy is a design choice but must ensure that:

- partial data is not presented as confirmed, publishable path information;
- failed transmissions are detected and either retried or reported to the user.

3.2 Functional Requirements

This section defines the functional requirements of the Best Bike Paths (BBP) system. Requirements are grouped according to the Product Functions F1–F6 defined in Section 2.2. Each requirement is labelled for traceability and later mapping to use cases.

F1 — User Management

- **RE-F1–1: Account Creation**

The system shall allow a person to create a registered user account, provided they supply the information required for authentication.

Supports UC1

- **RE-F1–2: User Authentication**

The system shall authenticate a user when valid credentials are provided.

Supports UC1

- **RE-F1–3: Access Control**

The system shall restrict access to trip recording and contribution functionalities to authenticated registered users only.

Supports UC1

- **RE-F1–4: Unregistered User Access**

The system shall allow unregistered users to request bike paths and view ranked results with no authentication.

Supports UC5

- **RE-F1–5: Session Termination**

The system shall allow a registered user to log out, after which no registered-only functionalities may be accessed.

Supports UC1

F2 — Trip Recording and Review

- **RE-F2–1: Start Trip Recording**

The system shall start recording a trip when a registered user initiates recording.

Supports UC1

- **RE-F2–2: Stop Trip Recording**

The system shall stop trip recording when a registered user ends the trip.

Supports UC1

- **RE-F2–3: GPS Data Acquisition**

The system shall acquire positional data (GPS coordinates) at regular intervals during an active trip.

Supports UC1.1

- **RE-F2–4: Sensor Data Acquisition**

The system shall acquire motion-related sensor data (accelerometer, gyroscope) during an active trip.

Supports UC1.1

- **RE-F2-5: Trip Statistics Computation**

The system shall compute trip statistics, including distance, duration and average speed, after recording ends.

Supports UC1

- **RE-F2-6: Weather Data Attachment**

The system shall retrieve and associate weather information with the recorded trip.

Supports UC1

- **RE-F2-7: Trip History Presentation**

The system shall display a list of previously recorded trips for each registered user.

Supports UC1

- **RE-F2-8: Trip Review**

The system shall display detailed information for any selected trip, including statistics and associated weather information.

Supports UC2

F3 — Manual Contribution and Maintenance of Path Information

- **RE-F3-1: Manual Path Selection**

The system shall allow registered users to select travelled path segments for reporting.

Supports UC3

- **RE-F3-2: Manual Issue Submission**

The system shall allow registered users to annotate selected path segments with status or obstacle information.

Supports UC3

- **RE-F3-3: Edit Existing Contributions**

The system shall allow registered users to update or remove their previously submitted path information.

Supports UC3

- **RE-F3-4: Publishable Information Storage**

The system shall store submitted or updated manual contributions as publishable path information.

Supports UC3

F4 — Automatic Detection and Validation of Issues

- **RE-F4-1: Issue Inference**

The system shall infer candidate issues from motion-related sensor data collected during a recorded trip.

Supports UC4

- **RE-F4-2: Candidate Issue Presentation**

The system shall present inferred candidate issues to the user for validation after the trip ends.

Supports UC4

- **RE-F4-3: User Validation Actions**

The system shall allow users to confirm, reject or refine candidate issues.

Supports UC4

- **RE-F4-4: Acceptance of Validated Issues**

The system shall store confirmed or refined issues as publishable information.

Supports UC4

- **RE-F4-5: Discarding Rejected Issues**

The system shall discard rejected issues so that they do not influence publishable path information.

Supports UC4

F5 — Path Search and Ranked Presentation

- **RE-F5-1: Path Query Input**

The system shall allow any user to provide an origin and a destination to request a bike path.

Supports UC5

- **RE-F5-2: Path Computation**

The system shall compute one or more candidate paths between the specified origin and destination, using external mapping or routing services when needed.

Supports UC5

- **RE-F5-3: Path Ranking**

The system shall compute a path score for each candidate path using segment conditions, quality indicators and path length.

Supports UC5

- **RE-F5-4: Path Presentation**

The system shall present ranked paths and their associated properties to the user.

Supports UC6

- **RE-F5-5: Path Details Display**

The system shall display detailed information for any selected path, including the conditions of its constituent segments.

Supports UC6

F6 — Aggregation and Maintenance of Path Knowledge

- **RE-F6-1: Aggregation of Issues**

The system shall aggregate issues from multiple users for each path segment.

Supports UC4

- **RE-F6-2: Freshness and Confirmation Priority**

The system shall resolve conflicts between issues by privileging fresher reports and those with more confirmations.

Supports UC4

- **RE-F6-3: SegmentStatus Update**

The system shall compute and update the SegmentStatus representing the current publishable condition of each path segment.

Supports UC4

- **RE-F6-4: Moderator Override**

The system shall allow moderators to override or correct aggregated segment statuses when necessary.

Supports UC4

- **RE-F6-5: Consistency of SegmentStatus**

The system shall ensure that at most one active SegmentStatus exists for each path segment at any given time.

Supports UC4

3.2.1 Traceability of Functional Requirements

The following tables provide traceability between functional requirements, use cases, goals, phenomena, and domain assumptions. Each table corresponds to one product function (F1-F6).

3.2.2 Traceability for F1 – User Management

Table 9: Traceability for F1 – User Management

RE	UC	Goals	Phenomena	DA	Description
RE-F1-1	UC1	G1, G2, G3	SP1	DA7	Enables account creation and identification
RE-F1-2	UC1	G1, G2, G3	SP1	DA7	Authenticates registered users
RE-F1-3	UC1,UC3, UC4	G1, G2, G3	SP1	DA6	Restricts access to protected functions
RE-F1-4	UC5, UC6	G4	SP6, SP7	–	Allows public access to path search
RE-F1-5	UC1	G1, G2, G3	SP1	–	Terminates authenticated sessions

3.2.3 Traceability for F2 – Trip Recording and Review

Table 10: Traceability for F2 – Trip Recording and Review

RE	UC	Goals	Phenomena	DA	Description
RE-F2-1	UC1	G1	SP2	DA1	Starts trip recording
RE-F2-2	UC1	G1	SP2	–	Stops trip recording
RE-F2-3	UC1.1	G1	SP3	DA1, DA2	Collects GPS data
RE-F2-4	UC1.1	G1, G3	SP3	DA1	Collects motion data
RE-F2-5	UC1	G1	–	–	Computes trip statistics
RE-F2-6	UC1	G1	SP9	DA3	Adds weather information
RE-F2-7	UC1	G1	SP8	–	Shows trip history
RE-F2-8	UC2	G1	SP8, SP9	–	Displays trip details

3.2.4 Traceability for F3 – Manual Contribution

Table 11: Traceability for F3 – Manual Contribution

RE	UC	Goals	Phenomena	DA	Description
RE-F3-1	UC3	G2	SP4	DA1	Selects travelled segments
RE-F3-2	UC3	G2	SP4	–	Submits path issues
RE-F3-3	UC3	G2	SP4	–	Updates or removes reports
RE-F3-4	UC3	G2, G6	SP4	–	Stores publishable data

3.2.5 Traceability for F4 – Issue Detection and Validation

Table 12: Traceability for F4 – Issue Detection and Validation

RE	UC	Goals	Phenomena	DA	Description
RE-F4-1	UC4	G3	SP3	DA1	Infers candidate issues
RE-F4-2	UC4	G3	SP5	–	Presents issues to user
RE-F4-3	UC4	G3, G6	SP5	–	Validates issues
RE-F4-4	UC4	G3, G6	SP5	–	Stores validated issues
RE-F4-5	UC4	G3	SP5	–	Discards rejected issues

3.2.6 Traceability for F5 – Path Search and Presentation

Table 13: Traceability for F5 – Path Search and Presentation

RE	UC	Goals	Phenomena	DA	Description
RE-F5-1	UC5	G4	SP6	–	Accepts path requests
RE-F5-2	UC5	G4	SP9	DA3, DA4	Computes candidate paths
RE-F5-3	UC5	G4, G6	SP7	–	Ranks paths
RE-F5-4	UC6	G4	SP7	–	Presents ranked paths
RE-F5-5	UC6	G4, G6	SP7	–	Shows path details

3.2.7 Traceability for F6 – Aggregation and Maintenance

Table 14: Traceability for F6 – Aggregation and Maintenance

RE	UC	Goals	Phenomena	DA	Description
RE-F6-1	UC4	G5, G6	SP4, SP5	–	Aggregates issue reports
RE-F6-2	UC4	G5	SP4, SP5	DA5	Resolves conflicts
RE-F6-3	UC4	G6	SP7	–	Updates segment status
RE-F6-4	UC4	G6	SP7	–	Allows moderator override
RE-F6-5	UC4	G6	–	–	Ensures consistency

3.3 Performance Requirements

The following performance requirements constrain the responsiveness and timing of BBP functionalities. Each requirement is grounded in the shared phenomena (SP), world phenomena (WP), and domain assumptions (DA) defined in earlier sections.

- **PR-1: GPS Sampling Frequency**

The system shall acquire GPS coordinates at intervals suitable for reconstructing the path taken by the user. The sampling rate should be constant even under poor signal conditions in order to prevent from losing trip path. (*Supports SP2, SP3; Relates to WP7; Depends on DA1, DA2*)

- **PR-2: Sensor Data Responsiveness**

The system shall process accelerometer and gyroscope data in near real time while a trip is being recorded. Delays in processing must be minimized if the inferred problems are to reflect true movement. (*Supports SP3; Relates to WP6; Depends on DA1*)

- **PR-3: Path Search Response Time**

The system shall compute and present path options within a few seconds under normal network conditions. Even if slower external routing services incur delays, those are well within the bounds of user waiting. (*Supports SP6, SP7; Depends on DA3*)

- **PR-4: Weather Data Retrieval Delay**

The system shall retrieve weather information with minimal delay following a request. More retries might be necessary when APIs externally to your own code are temporarily down or over rate limits. (*Supports SP9; Relates to WP4; Depends on DA3*)

- **PR-5: Aggregation Processing Time**

The system shall update SegmentStatus values promptly after receiving new manual or validated automatic contributions. Updates must be fast to ensure that newly identified problems are taken into account quickly in the path selection. (*Supports SP4, SP5, SP7; Relates to WP1; Depends on DA5*)

- **PR-6: Trip Summary Availability**

The system shall make trip statistics available immediately after the trip has ended, subject to network availability. The trip summary should be easily accessible as soon as you have completed making your review and there shouldn't be a need to manually refresh/reopen the app. (*Supports SP2, SP8; Relates to WP2; Depends on DA2*)

3.4 Design Constraints

The system shall comply with the following constraints derived from environmental factors, device capabilities, and domain assumptions.

- **DC-1: Device Sensor Availability**
The application will be supported on GPS, accelerometer and gyroscope enabled devices. Feature availability may be limited if devices have outdated or missing sensors. (*Relates to WP6, WP7; Depends on DA1*)
- **DC-2: External Service Dependency**
The system is expected to make use of weather and mapping services via an interface and is planned to interact with their ARN (availability resolution network). BBP should treat partial failures of these services gracefully by reverting to cached or stale data wherever suitable. (*Supports SP9; Relates to WP4; Depends on DA3, DA4*)
- **DC-3: Non-Safety-Critical Role**
The hazard warning or safety injecting shall not need any real-time from the system. BBP could be informative towards the environment of BBs, but would not deliver warnings in safety-critical context like collisions against obstacles. (*Relates to WP3; Depends on DA6*)
- **DC-4: Contribution Integrity**
The system will take the stance that users are contributing in good faith, meaning they have no malicious intent. This suggests that BBP lacks any feature to recognize deliberately false reports apart from aggregation logic. (*Relates to WP8; Depends on DA5*)

3.5 Software System Attributes

This section specifies the non-functional requirements that constrain the quality, robustness, and maintainability of BBP.

Reliability

- **NFR-REL-1: Connectivity Tolerance**
The system shall tolerate intermittent loss of network connectivity without interrupting the recording of local trip data. When offline, data needs to be securely buffered on the device until it can be transmitted. (*Supports SP2, SP3; Relates to WP2; Depends on DA2*)
- **NFR-REL-2: Status Consistency**
The system shall prevent inconsistencies such as duplicate or conflicting SegmentStatus entries for the same path segment. The system should apply timestamps and confirmation counts to resolve conflicting updates. (*Supports SP7; Relates to WP1; Depends on DA5*)

Availability

- **NFR-AV-1: Service Continuity**
Core functionalities (trip recording, path search) shall remain available provided that required device sensors and external services are reachable. When certain features (e.g., weather, geocoding) are temporarily unavailable the system should fail gracefully. (*Supports SP2, SP6; Relates to WP6, WP7; Depends on DA1, DA3*)

Security

- **NFR-SEC-1: Access Restriction**

Only logged-in users can post, edit, or validate published path contributions. The transmission of credentials needs to be secured against unauthorized interception. *(Supports SP4, SP5; Relates to WP8; Depends on DA7)*

- **NFR-SEC-2: Moderator Controls**

Top moderators for bypassing the cumulative status of a path will be limited to valid account(s) only. There should be moderation logs to keep everyone accountable. *(Supports SP7; Depends on DA7)*

Maintainability

- **NFR-MNT-1: Modular Structure**

The system shall isolate acquisition logic from aggregation and scoring logic to aid maintenance and extension. What modifies one module (scoring algorithm or the like) ought not affect independent parts of the system. *(Supports SP3, SP7; Depends on DA5)*

Portability

- **NFR-PORT-1: Device Compatibility**

The system shall operate on different mobile devices that support the sensors required by DA1. BBP must be seamless between different OS versions, assuming sensor APIs operate normally. *(Relates to WP6, WP7; Depends on DA1)*

3.6 Traceability Between Use Cases and Functional Requirements

Table 15: Traceability between Use Cases and Functional Requirements

Use Case	Functional Requirements
UC1 – Record Trip	RE-F2-1, RE-F2-2, RE-F2-5, RE-F2-7
UC1.1 – Acquire Sensor Data	RE-F2-3, RE-F2-4
UC2 – Review Trip	RE-F2-8
UC3 – Manual Path Contribution	RE-F3-1, RE-F3-2, RE-F3-3, RE-F3-4
UC4 – Confirm Detected Issues	RE-F4-1, RE-F4-2, RE-F4-3, RE-F4-4, RE-F4-5
UC5 – Search Path	RE-F5-1, RE-F5-2, RE-F5-3
UC6 – View Suggested Paths	RE-F5-4, RE-F5-5

4 Formal Analysis Using Alloy

This formal analysis aims to establish the logical consistency of the requirements of the creation of issues, validation and publishability of the issues regarding bike paths segments in the Best Bike Paths (BBP) system. Specifically, the analysis is devoted to the difference between manually reported and automatically identified issues and the regulations based on which they can be validated by the user confirmations. The model abstracts on purpose from some aspects of the system that are irrelevant concerning the properties analyzed. Specifically, it does not model users, trips, sensors, weather details, ranking functions, or user interfaces. Finally, time and quantity-related aspects, like freshness or number of confirmations, are not represented. These are abstracted away in order to limit the formal analysis to the logical consistency of issue validation and publishability constraints. It enables the model to be small and tractable while at the same time expressing the underlying constraints necessary for the satisfaction of Goals G2, G3, and G6.

```
module bbp_issue_validation

sig PathSegment {}

abstract sig Issue {
  segment: one PathSegment
}

sig ManualIssue extends Issue {}

sig AutomaticIssue extends Issue {}

sig Confirmation {
  issue: one AutomaticIssue
}

sig SegmentStatus {
  segment: one PathSegment,
  basedOn: some Issue
}

sig Validated in Issue {}
sig Rejected in Issue {}

fact AutomaticIssuesNeedConfirmation {
  all i: AutomaticIssue |
    i in Validated implies some c: Confirmation | c.issue = i
}

fact ValidationExclusivity {
  no i: Issue | i in Validated and i in Rejected
}

fact OnlyValidatedIssuesAffectSegmentStatus {
  all s: SegmentStatus |
    all i: s.basedOn | i in Validated
}

assert PublishabilityImpliesValidation {
  all s: SegmentStatus |
    all i: s.basedOn | i in Validated
}

assert AutomaticValidationRequiresConfirmation {
  all i: AutomaticIssue |
    i in Validated implies some c: Confirmation | c.issue = i
}
```

Figure 17: Alloy Code Part 1


```
pred ManualIssueValidatedWithoutConfirmation {  
  some i: ManualIssue |  
    i in Validated and  
    no c: Confirmation | c.issue = i  
}
```

Figure 18: Alloy Code Part 2

```
run {} for 5|
```

Line 55, Column 13 [modified]

Figure 19: Alloy Code Part 3

Executing "Run run\$1 for 5"

Actual scopes: 5 PathSegment, 5 Issue, 5 ManualIssue, 5 AutomaticIssue, 5 Confirmation, 5 SegmentStatus
Solver=sat4j Bitwidth=4 MaxSeq=5 SkolemDepth=1 Symmetry=20 Mode=batch
1569 vars. 135 primary vars. 2665 clauses. 4ms.
Instance found. Predicate is consistent. 3ms.

Figure 20: Execution of the code

The Alloy model describes a simplified understanding of the BBP domain based on the issue lifecycle and their contribution to the publishable information for bike path segments. This model also assigns a collection of core elements based on the identified concepts from the specified domain. They also have exactly one path segment identified and issues classified based on their type and origins. They also have confirmations to indicate the endorsement of the identified issues by the users. They have a description and analysis of the rules governing the validation and publishability based on a collection of constraints. These constraints also indicate that the automatically identified issues can only achieve validation when there is at least one confirmation. They also indicate that the issues reported by users are assumed to have been validated by default. They also restrict issues rejected or failing validation from building the publishable information. They have abstract descriptions of the publishable information for the path segments based on the segment statuses derived solely from the validated issues. They restrict the possibility of invalidated or rejected issues to affect the information provided to the users based on their choices. This Alloy model determines the formal verification to notice whether the specified constraints on the validation and publishability of issues have logical soundness based on the selected scope without any deviation.

```
check PublishabilityImpliesValidation for 5|
```

Line 55, Column 44 [modified]

Figure 21: Assertion and Check code part 1

Executing "Check PublishabilityImpliesValidation for 5"

Actual scopes: 5 PathSegment, 5 Issue, 5 ManualIssue, 5 AutomaticIssue, 5 Confirmation, 5 SegmentStatus
Solver=sat4j Bitwidth=4 MaxSeq=5 SkolemDepth=1 Symmetry=20 Mode=batch
1681 vars. 145 primary vars. 2854 clauses. 5ms.
No counterexample found. Assertion may be valid. 1ms.

Figure 22: Assertion and Check output part 1

check AutomaticValidationRequiresConfirmation for 5

Figure 23: Assertion and Check code part 2

Executing "Check AutomaticValidationRequiresConfirmation for 5"

Actual scopes: 5 PathSegment, 5 Issue, 5 ManualIssue, 5 AutomaticIssue, 5 Confirmation, 5 SegmentStatus
Solver=sat4j Bitwidth=4 MaxSeq=5 SkolemDepth=1 Symmetry=20 Mode=batch
1685 vars. 140 primary vars. 2864 clauses. 4ms.
No counterexample found. Assertion may be valid. 1ms.

Figure 24: Assertion and Check output part 2

To validate the correctness of the modeled constraints, we checked key properties through Alloy assertions. All checks were executed with a bounded scope of 5 instances for each signature. Alloy did not find counterexamples for the considered assertions, indicating that the modeled constraints enforce the intended requirements within the analyzed scope. Publishability implies validation. We verified that every issue contributing to publishable information (i.e., used by a segment status) must be validated. The check returned no counterexample found within the chosen scope. Automatic validation requires confirmation. We verified that an automatically detected issue cannot be validated unless at least one confirmation exists for that issue. The check returned no counterexample found within the chosen scope.

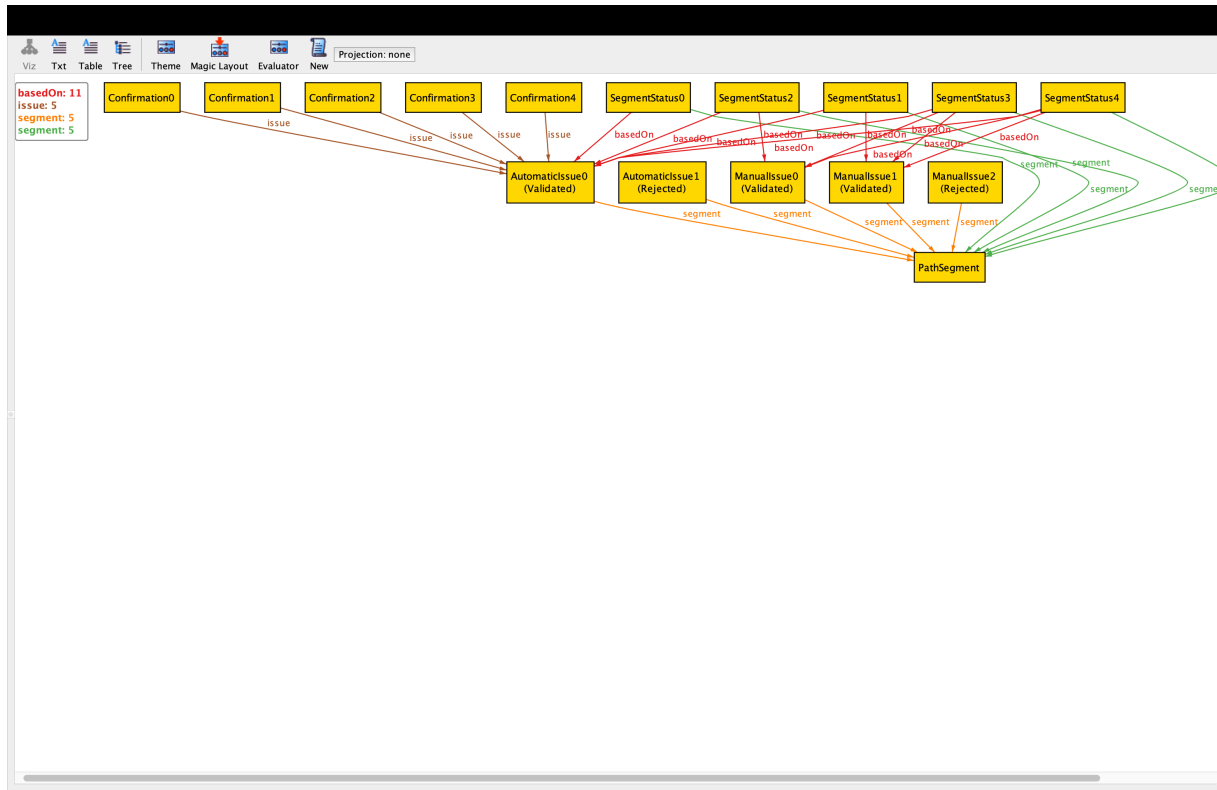


Figure 25: World 1 – Valid configuration of issues and segment statuses

This figure 25 illustrates a valid configuration of the Alloy model created by executing the model with no extra constraints. It includes both manually submitted and automatically identified issues. Automatically identified issues are checked only when they are tied with confirmations, whereas manually submitted issues can be checked directly. The segment statuses depend only on validated issues, which reinforces the belief that unvalidated or disqualified issues do not affect publishable paths.

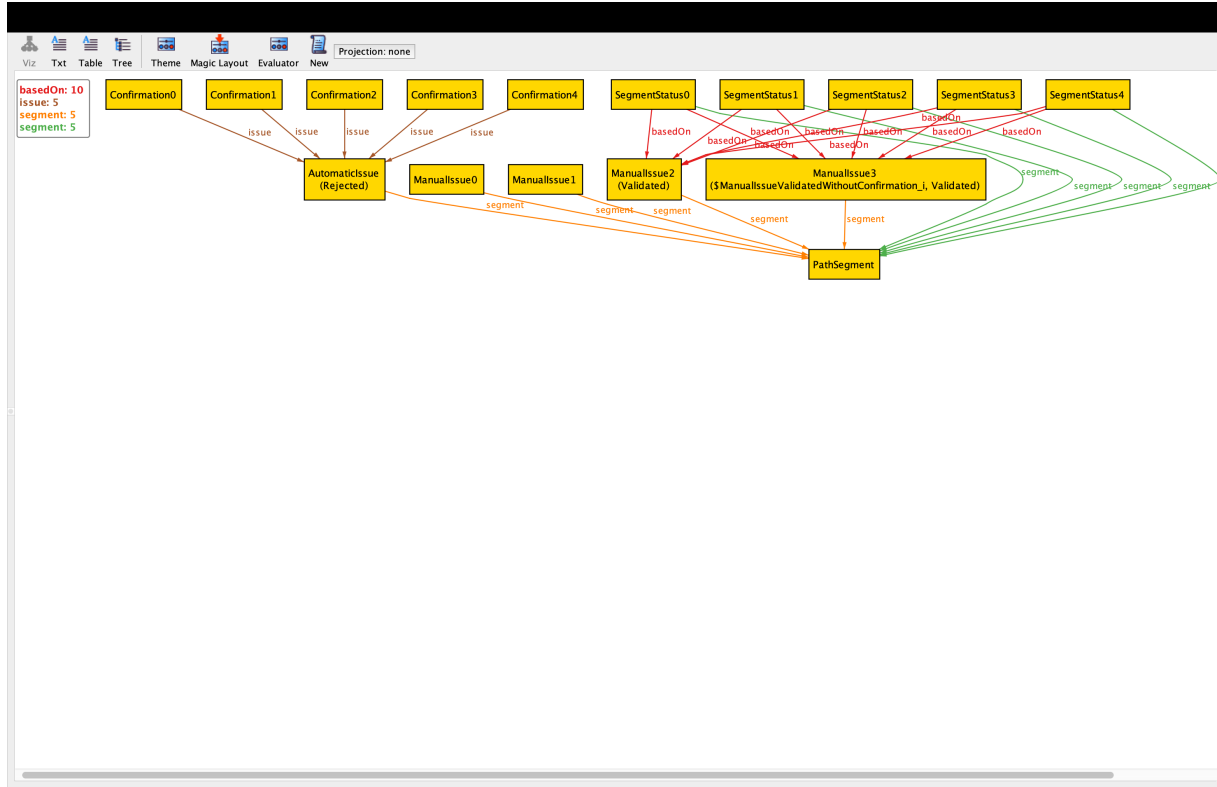


Figure 26: World 2 — Manual issue validated without confirmation

This figure 26 shows an instance generated by the predicate `ManualIssueValidatedWithoutConfirmation`. The instance contains a manually reported issue that is validated despite the absence of any confirmation. This execution demonstrates that the model correctly allows manual issues to be validated without confirmations, ensuring that validation rules are not over-constrained.

This formal verification confirms the logical correctness of the constraints regarding the validation of issues and publishability within the BBP system. Specifically, it checks that unparsed issues cannot be validated or affect publishable path information unless they are supported by user confirmations, and that only validated issues are reflected in segment statuses. These example executions also demonstrate that the model does not unduly constrain the system because manually entered issues can be validated without the need for confirmations. This verification largely confirms Goals G2, G3, and G6, as well as Use Case UC4.

5 Effort Spent

5.1 Ahmed Hassan

Task	Hours
Shared Sections Draft	2
Adding Section 1 from Draft	2
Fix Latex Errors and Repo Structure 1	4
Fix Latex Errors and Repo Structure 2	1.5
Adding 2.3 and 2.4.1 from Draft	3
Adding 2.1.2 text and diagrams	5
Adding 2.1.3 text and diagrams	5
Creating User Interface Mockups	7
Writing User Interface Description	4
Adding Hardware/Software/Communication inter- faces	4
Formal Analysis Using Alloy	8

5.2 Shuvro Ahmed

Task	Hours
Shared Sections Draft	2
Product perspective (Scenarios)	3
Use Case Diagram	4
Use Cases (UC 1 - 3)	3
Use Cases (UC 4 - 6)	2
Fix Latex Errors and Repo Structure 3	4
Sequence Diagrams (UC 1-3)	8
Sequence Diagrams (UC 4-6)	6
Requirement Mapping (G1 - G3)	4
Requirement Mapping (G3 - G6)	4
Fix Latex Errors and Repo Structure 4	2

5.3 Umme Humaira Himi

Task	Hours
Shared sections (Goals, Scope, Phenomena)	2
Goals and scenarios analysis for requirements	1
Product function structuring (F1-F6)	2
Functional requirements specification (F1-F3)	3
Functional requirements specification (F4-F6)	2.5
Functional requirement refinement	0.5
Requirement alignment with use cases	2.5
Performance requirements	1.5
Design constraints	2
Software system attributes	1.5
Traceability tables (F1-F3)	2
Traceability tables (F4-F6)	1
UC, goal, phenomena, and assumption cross-check	2
Final requirements consistency check	1.5

References

The following tools and resources were used in the preparation of this document:

- **draw.io** (<https://www.drawio.com>): Used to create use case diagrams.
- **Alloy Analyzer** (<https://alloytools.org>): Used for formal analysis and Alloy models.
- **Mermaid Live Editor** (<https://mermaid.live/>): Used to create domain class diagrams and state diagrams.